



Influence of Mississippi and Atchafalaya River plume in the winter coastal cooling of the Northwestern Gulf of Mexico

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ABSTRACT

Sea surface temperature (SST) variability was analyzed in the Northwestern Gulf of Mexico over the period 1982–2018. This analysis was carried out by the combination of reanalysis databases (OISST_{1/4} and CFSR) and a hydrodynamic model (ROMS). SST warming is detected throughout the year with the exception of the Louisiana-Texas shelf where a cooling is observed during winter months (November–February). The existence of a different SST trend near coast with respect to open ocean is related to Mississippi and Atchafalaya River plumes. The bigger differences, between plume area cooling and ocean warming, are detected under the plume reinforcement driven by an increment in the freshwater discharge. The plume reinforcement increases the stratification and reduces the vertical mixing, favoring the atmospheric cooling of coastal SST during winter.

1. Introduction

Sea Surface Temperature (SST) has shown a strong warming trend since the 1980s (IPCC, 2013) due to the increase in the heat content of the Earth, which has been mainly absorbed by oceans (Church et al., 2011; Levitus et al., 2012). Although SST warming was the dominant trend in all basins, it is important to take into account that variations are highly dependent on time and spatial scales. Thus, the sign of the trend can change depending on the decades considered (deCastro et al., 2009; Gouretski et al., 2012) and on the season of the year examined (Costoya et al., 2015). Regarding spatial scales, warming rates change from basin to basin (Levitus et al., 2012) being variations more marked at regional scales.

Trends in coastal areas have also shown high spatial heterogeneity over the last three decades (Lima and Wetthey, 2012). Coastal areas can be subjected to different forcing favoring a different warming pattern when compared with open water. The main factors contributing to the different behavior are: freshwater inputs (e.g. Howden and Murtugudde, 2001; Vizu and Cook, 2010; Park et al., 2011; Materia et al., 2012; White and Toumi, 2014; Costoya et al., 2016, 2017), upwelling (e.g. Relvas et al., 2009; Santos et al., 2012a, 2012b, 2012c, 2015; Varela et al., 2015) or the presence of transitional water bodies (e.g. Kozlov et al., 2014). These forcing in combination with global warming can provoke a different SST response depending on the coastal

area and the season considered.

Regarding freshwater inputs, some studies have found that river discharge can modulate or even reverse SST trends in close areas. Vizu and Cook (2010) noticed a higher warming during austral summer in the area around Amazonas and Orinoco mouths. Materia et al. (2012) and Park et al. (2011) found the same effect due to Congo River and Changjiang River discharges during summer. However, Howden and Murtugudde (2001) detected a cooling trend in the Bay of Bengal provoked by river runoff by means of numerical simulations and White and Toumi (2014) also observed a cooling effect in localized regions of Congo River mouth using a regional ocean model. Other studies (Seo et al., 2009; Behara and Vinayachandran, 2016) noticed a higher warming (cooling) during summer (winter) in the Bay of Bengal also by means of model simulations. Finally, Costoya et al. (2016) found a SST cooling along the French continental shelf during winter months due to the river plume formed by Loire and Gironde Rivers. Therefore, the role of river inputs in modifying the thermal characteristics of the adjacent water varies depending on the region and the season considered. For these reasons, it is necessary to know the influence of the most important fluvial systems on the temperature patterns of seawater. The knowledge of the systems able to modify ocean temperature trends has a remarkable importance in the actual context of global warming, since it is a well-known fact that changes in sea temperature are considered as the most pervasive and severe impact in coastal ecosystems (Lima

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and Wetthey, 2012). In this sense, despite the numerous studies on the plume formed by the Mississippi and Atchafalaya River discharges, there is not an in depth analysis on its influence on SST and its variations along the last decades. The knowledge of the influence of this plume system on the thermohaline properties of the seawater and their evolution, can help to better understand the role of these systems as modulators of global warming. In addition, this information can also help to understand the effects of other minor river systems subject to similar conditions and where the data acquisition is more limited.

The aim of this study is to analyze the SST trend in the Northwestern part of the Gulf of Mexico from 1982 to 2018 as well as the influence of the plume formed by the Mississippi and Atchafalaya River system on it. In addition, factors able to determine the plume dynamic and, therefore, to modify coastal SST, will be assessed to determine their influence. The Regional Ocean Modeling System (ROMS) will be used to evaluate the plume development as well as changes on it, and the $\frac{1}{4}^\circ$ daily Optimum Interpolation SST (OISST) database will be used to quantify variations in SST. Section 2 shows the databases used to carry out the present study, as well as the methodology developed to analyze SST variations. Results are described in Section 3 and discussed in Section 4, where the main conclusions are also shown.

2. Study area, databases and methodology

2.1. Study area

The Mississippi River is one of the most important rivers in the world in terms of river discharge and environmental settings. In fact, Mississippi is the longest river in North America (6300 km), with an average discharge of about $20,000 \text{ m}^3 \text{ s}^{-1}$, draining approximately 41% of the continental United States (Walker et al., 2005; Zhang et al., 2012). The Mississippi River splits into two distributaries upstream of the Tarbert Landing station. The Old River outflow channel, carrying one-third of the upper Mississippi River flow, merges with the Red River, near Simmesport, to form the Atchafalaya River. The Mississippi-Atchafalaya (hereafter referred to as MA) is the seventh largest river system by water discharge ($> 5 \times 10^{11} \text{ m}^3 \text{ year}^{-1}$) in the world (Milliman and Meade, 1983; Allison et al., 2012; Marta-Almeida et al., 2013). This river system provides, at the Louisiana-Texas shelf, around 90% of the river discharges into the Gulf of Mexico (Androulidakis et al., 2015). The Louisiana-Texas shelf is located in the Northwestern area of the Gulf of Mexico (Fig. 1a). The width of the shelf is around 200 km in the area close to the Atchafalaya bay and $< 50 \text{ km}$ near the Mississippi River mouth (Zhang et al., 2012). The Louisiana-Texas shelf is an economically important coastal region due to extensive offshore

oil and gas activities, commercial and fishing, as well as coastal tourism (Zhang and Hetland, 2012).

The MA River system provides an important input of freshwater, sediment, nutrients, and pollutants, causing a major impact on the parameters controlling the continental shelf oceanography in the Northern Gulf of Mexico. On the one hand, the supply of nutrients enhances the phytoplankton and zooplankton productivity. This fact makes the Louisiana-Texas shelf a very productive fishery area, with around 30% of total U.S. catchment, being also the spawning area of several essential species (Walker, 1996). On the other hand, this high productivity presents some negative consequences such as a large zone of hypoxia generated during summer months leading to eutrophication (Walker, 1996).

For the reasons above mentioned, it is of great importance tracking the MA plume development and its impact on the most relevant oceanic parameters. The impact of MA River plume on different variables was analyzed by several authors through numerical modeling (Xu et al., 2011; Hetland and Dimarco, 2012; Zhang et al., 2012; Marta-Almeida et al., 2013; Rong et al., 2014; Androulidakis et al., 2015), satellite imagery (Miller and McKee, 2004; Salisbury et al., 2004; Walker et al., 2005; Hu et al., 2005; Shi and Wang, 2009), in situ data (Corbett et al., 2004; Allison et al., 2012; Joung and Shiller, 2014) or with the combination of these methodologies (Kolker et al., 2014). The majority of these studies are focused on the displacement of the plume along the Louisiana-Texas shelf under different conditions and on the analysis of the plume influence on some parameters related to primary productivity. In general terms, the MA plume tends to be displaced to the west due to the buoyant forcing and the predominant east component of winds, especially during non-summer months (Schiller et al., 2011). This westward transport is reduced during summer due to the predominant, although weak, upwelling favorable winds (Walker et al., 2005; Marta-Almeida et al., 2013). The Loop Current and eddies can also affect the river plume in the vicinity of the Mississippi delta especially, under conditions favoring the offshore movement of the plume (Walker et al., 2005).

2.2. Sea surface temperature data

SST data were obtained from the daily Optimum Interpolation Sea Surface Temperature database (OISST_{1/4}, version 2, Reynolds et al., 2007). It combines in situ data from ships and buoys and satellite data to construct a regular global grid by means of a special form of kriging called Optimal Interpolation. This procedure retains large-scale correlation structures and allows assimilating very sparse data coverage (Reynolds, 2009; Reynolds and Chelton, 2010). Regarding satellite

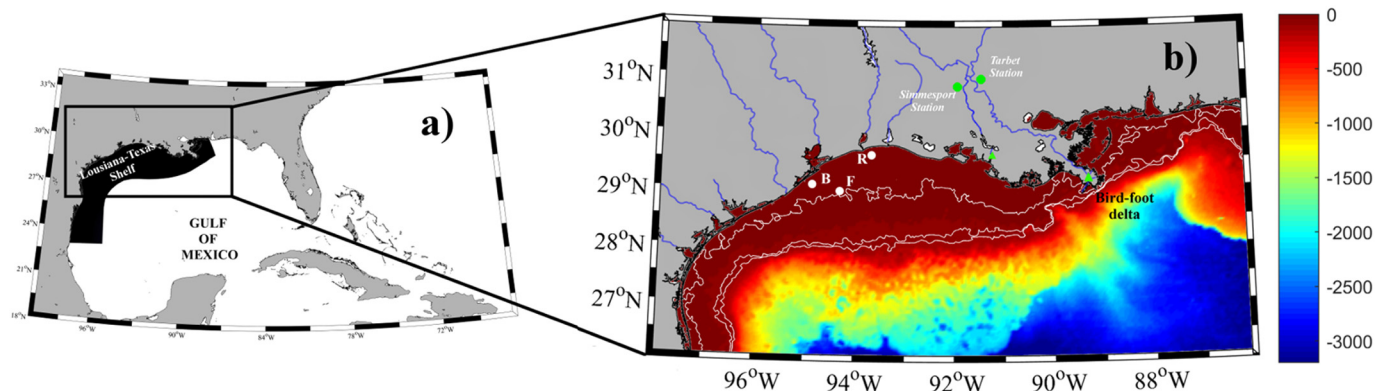


Fig. 1. a) Location of the region under scope. Black area represents ROMS model domain. b) Bathymetry of the region under scope. Green dots mark the location where river discharge was sampled. Green triangles mark the location of the main mouth of Atchafalaya and Mississippi Rivers. White dots mark the location of the buoy TABS used in the study. The nine rivers included in ROMS model are, from west to east: Nueces, San Antonio, Lavaca, Brazos, Trinity, Sabine, Calcasieu, Atchafalaya, Mississippi. White contour lines represent 20, 75 and 100 m isobaths. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

data, only data from the Advanced Very High Resolution Radiometer (AVHRR) sensor were used because it provides an uninterrupted SST data since March 1981 (Casey et al., 2010) with a spatial resolution of $0.25^\circ \times 0.25^\circ$. Daily files for the northwestern area of the Gulf of Mexico ($86\text{--}98^\circ\text{W}$, $26\text{--}31^\circ\text{N}$) were retrieved from the National Oceanic and Atmospheric Administration (NOAA) website (<http://www.ndc.noaa.gov/sst/>) over the period 1982–2018. Daily SST values were averaged at monthly scale. Monthly anomalies were obtained by subtracting for each grid point and month, the mean monthly SST over the whole period (e.g. mean of all Januaries from 1982 to 2018) from the monthly SST value (e.g. January 1982). Finally, SST trends were calculated using monthly anomalies and were assumed to be linear.

SST anomalies at coastal and oceanic locations were further analyzed by means of the empirical orthogonal functions (EOFs) method.

Daily SST data were also obtained from the Multi-scale Ultra-high Resolution (MUR) SST Analysis, which provides global SST data with a spatial resolution of 0.01° from 2002 to present. Although the time period available from this database is too short to provide consistent temperature trends, it allows to test, by comparison, if the spatial resolution of OISST adequately represents the main features of the coastal area. Daily SST values were averaged at monthly scale and monthly anomalies were calculated to remove the seasonal cycle.

2.3. Regional Ocean Modeling System (ROMS) data

Water salinity, temperature and velocity data were retrieved from ROMS model. This model domain covers the Louisiana-Texas shelf (Fig. 1a) with a spatial resolution ranging from approximately 600 m near the coastline to 4 km along the offshore open boundary. The number of vertical layers was 30 (Zhang et al., 2012). The model was run with realistic forcing and point sources. Surface fluxes (momentum and tracer) were obtained from ERA Interim. Streamflow was included as point sources and nine rivers across Texas and Louisiana, including Mississippi and Atchafalaya Rivers, were included in the model (see Fig. 1). The daily streamflow data were obtained from the US Army Corps of Engineers and US Geological Survey. Tides are not included in the model as they are known to be small in the region (DiMarco and Reid, 1998; Salisbury et al., 2004). The model was nested into global ocean general circulation models (OGCM), namely Global HYCOM (1993–2012) and Global Mercator (2013–2018). The hindcast model was run for the 1993–2018 period with a spin-up period of one year. Data from 1994 to 2018 were used in this study.

Temperature and salinity data were used to analyze the stratification of the water column over the period 1994–2018. The buoyancy frequency (N^2 , also called Brunt-Väisälä frequency squared), which represents the stratification strength at a specified depth (z), was defined as follows:

$$N^2 = -\frac{g}{\rho_z} \frac{\Delta\rho}{\Delta z} \quad (1)$$

where g is the gravity constant, ρ_z is the average water density (kg m^{-3}) and the term $\frac{\Delta\rho}{\Delta z}$ accounts for the potential density gradient from surface to the specified depth z . Brunt-Väisälä frequency was calculated for the all depths available. Then, Maximum Brunt-Väisälä values along the water column were obtained for each grid and time (at a monthly scale).

Changes in the freshwater transport were analyzed by means of salinity and velocity data, following a procedure based on Schiller et al. (2011):

$$\Delta\text{FW}_{\text{transport}}^{i,j} = \left(\frac{1}{N_a} \sum_{k=1}^{N_a} U_k^{i,j} S_n^{i,j} - \frac{1}{N_b} \sum_{k=1}^{N_b} U_k^{i,j} S_n^{i,j} \right) \quad (2)$$

where i,j refer to the pixel, k to the time instant and, N_a and N_b to the number of time instants in the interval 1994–2006 and 2006–2018, respectively. U is the zonal component of the velocity and S_n is a proxy

of the freshwater content calculated as

$$S_n^{i,j} = \left(\frac{S_{\text{ref}} - S_k^{i,j}}{S_{\text{ref}}} \right)$$

where $S_{\text{ref}} = 37$ is a reference salinity estimated from the maximum salinity at ocean locations.

2.4. Texas Automated Buoy System (TABS) data

In situ temperature and salinity data derived from the TABS were used to assess the performance of the OISST_{1/4} database and ROMS model. TABS is a coastal network of moored buoys along the Texas coast maintained by the Geochemical and Environmental Research Group at Texas A&M University (<http://tabs.gerg.tamu.edu/tglo/>). Temperature and salinity data from three buoys (B, F and R, see Fig. 1b) located at approximately 2 m below sea surface were considered in this study. The Buoy F provides data since 1996 and the Buoys B and R since 1999. Mean daily temperature and salinity values were obtained from the buoys. The closest pixel of the OISST_{1/4} and ROMS to each buoy was considered and the selected period was the maximum available for each buoy. Daily values were averaged for each month and monthly anomalies were calculated in order to remove the seasonal component and to prevent artificial correlations. Finally, Spearman rank correlation coefficient (hereafter referred to as correlation coefficient) was used to analyze the suitability of OISST_{1/4} data to represent real SST and of ROMS data to represent real temperature and salinity. Spearman rank correlation was selected to analyze the significance of trends and correlations due to its robustness to deviations from linearity and its resistance to the influence of outliers.

2.5. Heat flux, air temperature and wind data

Surface heat flux (shortwave, longwave, latent and sensible), air temperature and wind data were analyzed due to their key role in SST patterns. These variables were obtained from the Climate Forecast System Reanalysis (CFSR) developed by NOAA's National Centers for Environmental Prediction (Saha et al., 2010) (<http://rda.ucar.edu/pub/cfsr.html>) for the period 1982–2018, which allows cover the total extent of the study. In addition, this database provides a spatial resolution of $0.3^\circ \times 0.3^\circ$ from January 1982 to April 2011 and $0.2^\circ \times 0.2^\circ$ thereafter. Data from the latter period were interpolated on a $0.3^\circ \times 0.3^\circ$ grid to use a common resolution over the whole period. This spatial resolution is fine enough to analyze the main processes occurring at coastal areas. Wind data measured at a reference height of 10 m and at 6-h intervals were considered in order not to filter daily variations in wind patterns. Air temperature and surface heat flux data measured at 2 m were averaged at monthly scale.

The net heat flux (Q_T) through the ocean surface was evaluated following Eq. (3):

$$Q_T = Q_{\text{SW}} + Q_{\text{LW}} + Q_S + Q_L \quad (3)$$

where Q_{SW} is the shortwave flux, Q_{LW} is the longwave flux, Q_S is the sensible heat flux and Q_L is the latent heat flux. According to the selected sign criterion, negative heat fluxes imply that ocean is losing heat whereas positive heat fluxes indicate that ocean is gaining heat.

2.6. River discharge data

Daily runoff data for Mississippi and Atchafalaya Rivers were obtained from gauging stations operated by the US Army Corps of Engineers (<http://rivergages.mvr.usace.army.mil/WaterControl/new/layout.cfm>). In particular, river discharge data were retrieved at the Tarbert Landing station for Mississippi River and at Simmesport for Atchafalaya River (Fig. 1b, green dots). Daily data over the period 1982–2018 were averaged at monthly scale to analyze the river

Table 1

Temperature correlation between OISST_{1/4} and ROMS databases and TAB buoys, and the associated RMSE (root mean square error). All values are significant at 99%.

	Buoy B		Buoy F		Buoy R	
	Correlation	RMSE	Correlation	RMSE	Correlation	RMSE
OISST _{1/4}	0.77	1.47	0.83	1.41	0.84	1.74
ROMS	0.86	1.15	0.78	0.89	0.87	1.22

discharge variability and its influence on SST patterns.

2.7. The AMO teleconnection index

The Atlantic Multidecadal Oscillation (AMO) is a mode representing the multidecadal variability of the SST over the North Atlantic Ocean. The AMO index is calculated averaging the SST anomalies in the Atlantic Ocean north of the Equator (Enfield et al., 2001). It was obtained from the Climate Prediction Center at the National Center of Environmental Prediction (<http://www.cpc.ncep.noaa.gov>) at a monthly scale for the period 1982–2018.

3. Results

3.1. Validation

The capability of OISST_{1/4} and ROMS datasets to analyze SST variability was evaluated by means of correlations with TABS buoys for the whole period available following the procedure explained in Section 2.4, where the seasonal cycle was removed for all databases in order to avoid artificial correlations (Table 1). Both OISST and ROMS data sets showed high correlation coefficients (around 0.80 with a significance > 99%) with the three buoys closest to the Atchafalaya and Mississippi mouths (R, F and B). The Root Mean Square Errors (RMSE) show values around 1.5 °C for the OISST and 1.1 °C for the ROMS datasets, which are similar to previous analysis (Androulidakis and Kourafalou, 2013).

In addition, salinity from ROMS model was also compared with TABS buoys following the same approach done for SST (Table 2). High Spearman rank correlation values (> 0.8) were obtained for buoys B and R. These correlation values are similar to the ones found for Androulidakis and Kourafalou (2013). RMSE values around 2 were obtained for buoys B and R and slightly higher for buoy F. Therefore, attending to Tables 1 and 2, the accuracy of OISST_{1/4} and ROMS datasets to reproduce SST and salinity variability allows analyzing other regions where in situ measurements were not available. Finally, SST data from the OISST_{1/4} database was also compared with the SST from the high resolution MURSST database over the common time period (2002–2018) to analyze if the OISST_{1/4} spatial resolution could suppose a source of uncertainty near coast. Results show that both databases adequately reproduce the main features of the studied area (Supporting Information S1). Therefore, the OISST_{1/4} database was selected to analyze SST patterns and their variability in the area under study allowing to provide robust trends over > 30 years. In fact, OISST_{1/4} was successfully used in several coastal studies related to temperature

Table 2

Salinity correlation between ROMS model and TAB buoys, and the associated RMSE (root mean square error). All values are significant at 99%.

	Buoy B		Buoy F		Buoy R	
	Correlation	RMSE	Correlation	RMSE	Correlation	RMSE
ROMS	0.84	2.09	0.64	3.37	0.84	1.93

trends in other locations (Lima and Wetthey, 2012; Costoya et al., 2016; Varela et al., 2018).

3.2. Sea surface temperature

SST trends in the Northwestern Gulf of Mexico were calculated at monthly scale over the period 1982–2018 using the OISST_{1/4} database (Fig. 2). Overall, warming trend is dominant for all months with maximum values of 0.5 °C decade⁻¹. However, it stresses the presence of a cooling strip along the Louisiana-Texas shelf from November to February that contrasts with the warming observed at open water. The difference in the warming rate between coastal and oceanic locations is even more striking in December and January, when a higher significant cooling trend was observed. SST trends were averaged over the period November to February (NDJF SST) (Fig. 3), which is characterized by a clear contrast between cooling and warming areas as shown in the previous figure. A significant cooling trend was detected along the entire Louisiana-Texas shelf with values around -0.5 °C decade⁻¹. It should be highlighted that the cooling was widespread to the west of the bird-foot delta but not eastward.

In order to analyze the different warming patterns between coastal and ocean locations, both regions were delimited (see mini-map in Fig. 4). Blue points mark the near coast locations where the highest cooling was observed from November to February. Magenta dots mark an ocean band where the maximum warming was observed over the same months. Therefore, coastal and oceanic regions were delimited attending to the different SST trend observed. A thin band among them, where there is not a clear trend, and the ocean points far from the coast were not considered in this analysis. Points with similar trends (positive or negative) were spatially averaged at monthly scale. Both areas have practically the same warming rate (around 0.25 °C decade⁻¹) from April to October. Although the trends are not exactly equal in March, they are positive both for coast and ocean. In contrast, large differences can be observed from November to February with the coastal area cooling (blue line) and the adjacent oceanic area (magenta line) warming.

The interannual variability of coastal and oceanic SST is shown in Fig. 5a for the season under interest (from November to February, NDJF SST from now on) over the period 1982–2018. In general, coastal SST (blue line) is lower than 20 °C and presents a negative trend and ocean SST (red line) is higher than 21 °C and with a positive trend. The difference between both signals (Fig. 5b) can be observed to increase progressively throughout most of the studied period with SST differences higher than 4 °C from 2007. This increment is even more marked over the last decade (~ 0.2 °C year⁻¹).

3.3. Air-sea heat exchange and air temperature

The role of the atmospheric forcing on the different warming pattern between coastal and oceanic locations was evaluated throughout the analysis of the air-sea heat exchange (Fig. 6) and throughout the changes in the air temperature (Fig. 7) from November to February over the period 1982–2018. Ocean loses heat (negative sign) in the whole area under scope (Fig. 6a). However, the average rate is larger at ocean locations (ranging from 150 to 200 W m⁻²) than at coastal locations along the Louisiana-Texas shelf (ranging from 75 to 125 W m⁻²), in response to the different temperature between coastal and oceanic locations (heat loss is more intense in the areas where SST is also higher). Regarding trends (Fig. 6b), a different pattern was observed along the Louisiana-Texas shelf in comparison with ocean locations, which remarks the different behavior between both areas, as previously described for SST trends. In this case, positive trends were detected along the entire Louisiana-Texas shelf and negative trends prevailed in the rest of the area under scope, in response to the different temperature evolution detected in both areas. Note that positive trends mean that the rate of heat loss was reduced over the studied period and less energy is

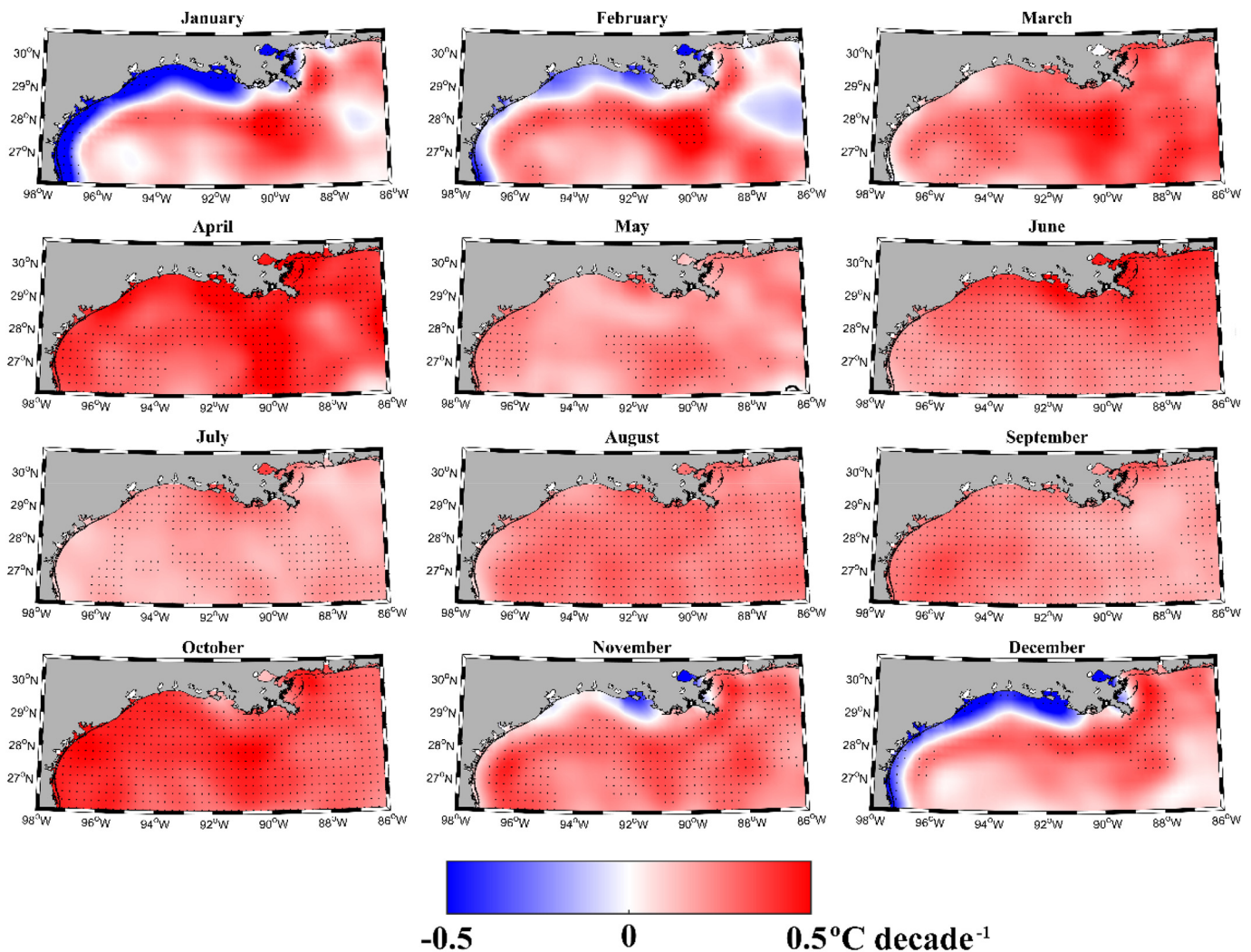


Fig. 2. Monthly SST trends ($^{\circ}\text{C decade}^{-1}$) along the Louisiana-Texas shelf over the period 1982–2018. Trends are calculated by means of the OISST_{1/4} database. Black dots represent the grid points with a significance higher than 95%.

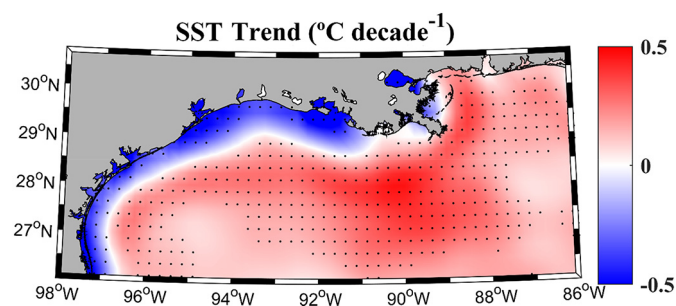


Fig. 3. November–February SST trend ($^{\circ}\text{C decade}^{-1}$) calculated over the period 1982–2018. Black dots represent grid points with a significance higher than 95%.

transferred from the ocean to the air. The role of the atmospheric forcing was also analyzed throughout the air temperature trend (Fig. 7). There are not differences between coastal and oceanic trends. The whole area, including both land and ocean, shows a similar increase (between 0.1 and $0.3^{\circ}\text{C decade}^{-1}$) in the air temperature for the period under study.

3.4. Thermohaline variables

The evolution of thermohaline properties (temperature, salinity and

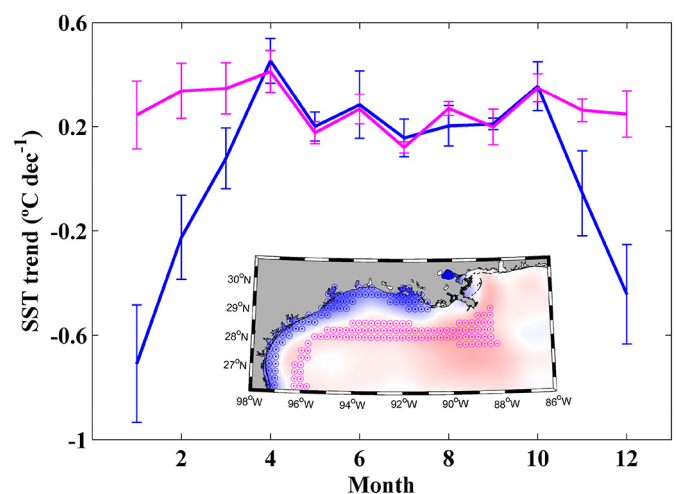


Fig. 4. Monthly trend and its associated variance (bars) for oceanic (magenta line) and coastal (blue line) areas. The mini-map shows the averaged points selected to calculate monthly trends at coastal (blue) and oceanic (magenta) areas. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

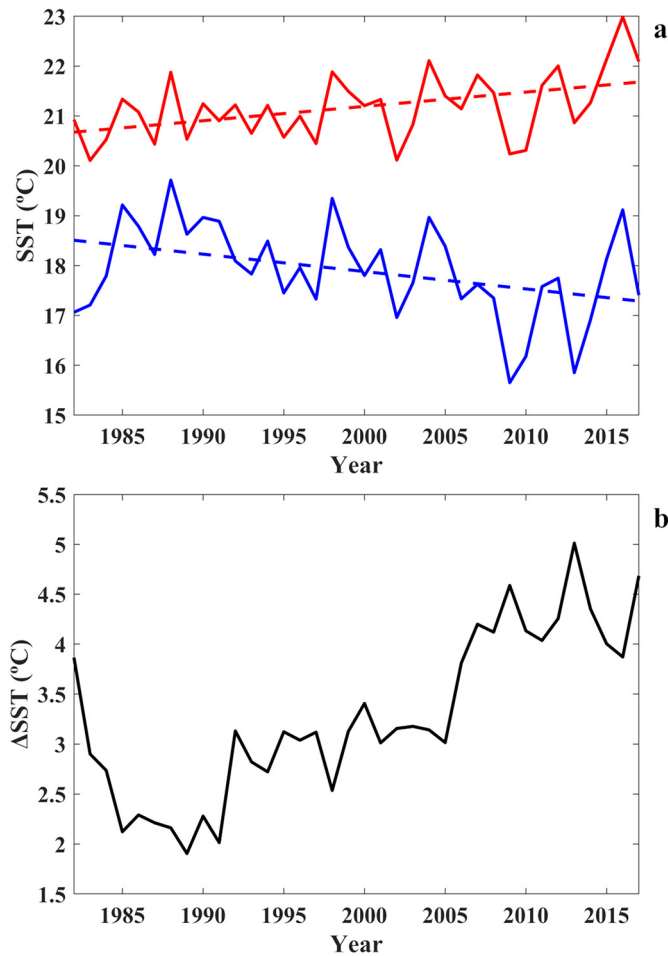


Fig. 5. (a) Interannual NDJF SST variability in oceanic (red line) and coastal (blue line) areas obtained from the OISST_{1/4} database. Dashed lines show the linear trend. Both trends have a significance higher than 95%. (b) Interannual NDJF SST difference between both regions over the period 1982–2018 obtained from the OISST_{1/4} database. NDJF Δ SST = NDJF SST_{ocean} – NDJF SST_{coast} . The oceanic and coastal areas were defined in Fig. 4. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

density), analyzed using ROMS data, shows the role of the ocean dynamics in the coastal cooling. A similar NDJF SST trend was obtained from 1994 to 2018 (Fig. 8a). A decreasing temperature trend was detected near the coast; whereas, the oceanic water tended to warm. This pattern was also detected using OISST_{1/4} dataset over the period 1994–2018 (results not shown), which corroborates that cooling-warming pattern is present regardless the time period or database considered. Fig. 8b shows a salinity decrease at surface layer in the coastal area and a salinity increase in the adjacent ocean. Regarding density, a similar pattern than the salinity one was detected (Fig. 8c), which, as expected, indicates that this variable is dominated by the salinity, especially in the coastal area. In addition, the evolution of the NDJF salinity and temperature along the water column in the cooling area was analyzed by means of the vertical profiles obtained from ROMS. Vertical profiles result from spatially averaging coastal points where cooling was observed. This analysis was made considering the first 20 m of the water column due to the shallowness of the coastal region analyzed (see Fig. 1b). In terms of mean values, salinity (Fig. 9a) and temperature (Fig. 9b) increases with depth, showing, in the last case, a thermal inversion. Changes in the evolution of NDJF salinity and temperature were detected subtracting the mean values for the period (1994–2006) from the mean values for the period (2006–2018). In this

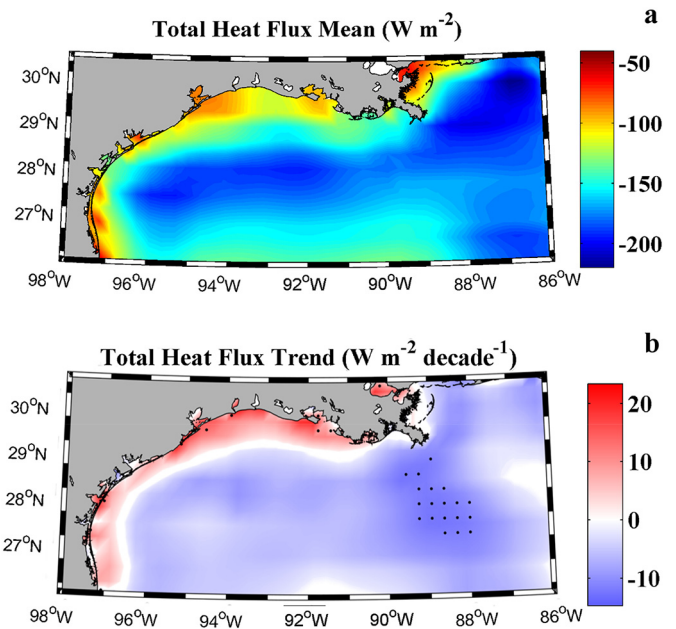


Fig. 6. (a) Mean heat flux ($W m^{-2}$) and (b) trend ($W m^{-2} decade^{-1}$) calculated from November to February over the period 1982–2018. Black dots represent grid points with significance higher than 95%. A negative (positive) value implies that ocean is losing (gaining) heat.

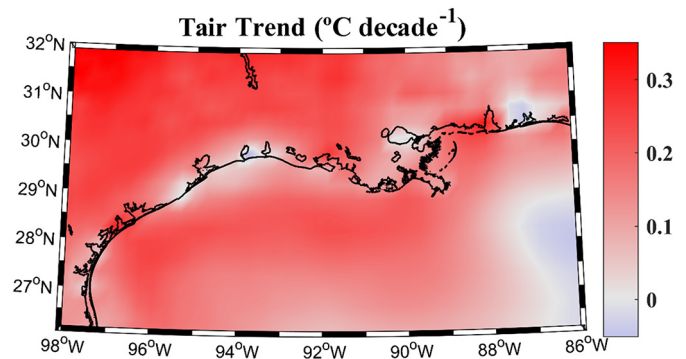


Fig. 7. Air temperature trend ($^{\circ}C decade^{-1}$) from November to February over the period 1982–2018. Air temperature data measured at 2 m were obtained from the CFSR database.

way, negative (positive) values indicate the decrease (increase) of these variables. In general terms, a decrease in salinity along the water column was observed, which is especially remarkable in the first 10 m, with maximum values around -0.55 (Fig. 9c). Respect to temperature, a decrease around -0.35 $^{\circ}C$ was maintained in the first 10 m of the water column. At deeper layers the temperature decrease is lower reaching values close to -0.15 $^{\circ}C$ (Fig. 9d).

Attending to these results, it is clear that coastal and oceanic locations present different thermohaline evolutions. In this context, different drivers can affect surface coastal water. As it was mentioned in the Introduction section, the Louisiana-Texas shelf is highly influenced by freshwater inputs, mainly from Mississippi and Atchafalaya Rivers. The river plume formed by these rivers can affect significantly the salinity, temperature and density patterns of seawater. In order to test if the area occupied by the MA plume is related with the cooling area, the plume formed by these rivers from November to February was evaluated by means of salinity data from ROMS (the main natural tracer of river plume) (Fig. 10). It can be observed as the river plume, characterized by low salinity values (below 34) (Androulidakis and Kourafalou, 2013), tended to expand westward and its presence was

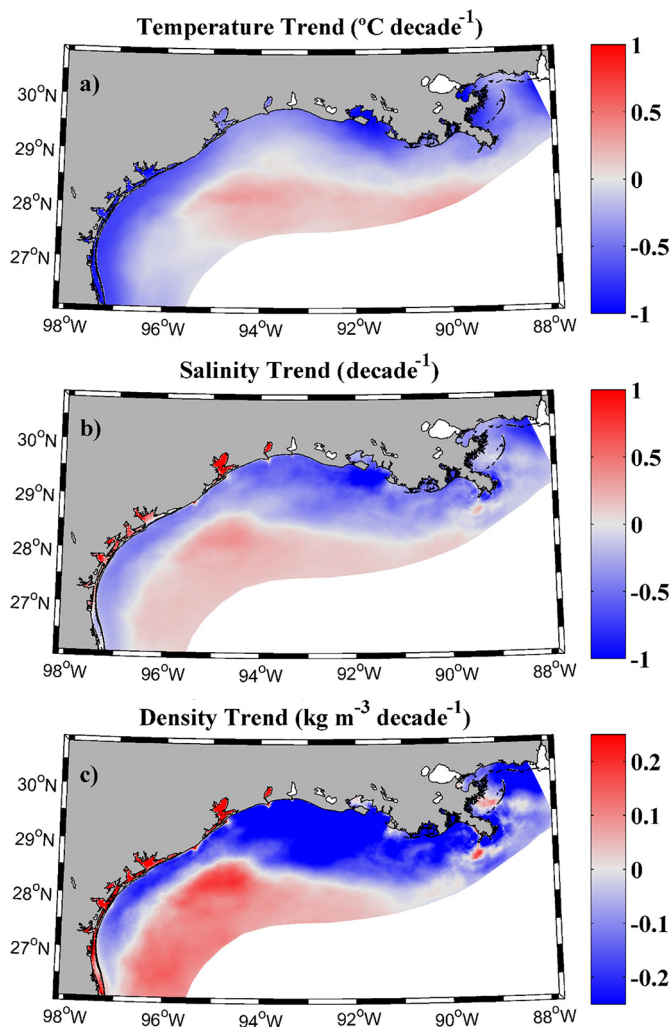


Fig. 8. Trends in near surface (a) temperature ($^{\circ}\text{C decade}^{-1}$), (b) salinity (decade^{-1}), and (c) density ($\text{kg m}^{-3} \text{decade}^{-1}$), calculated from November to February over the period 1994–2018.

scarce eastward of the bird-foot delta where Mississippi River flows into, coinciding with the cooling area (Fig. 3). This pattern indicates that the river plume plays a key role in the temperature patterns observed. In addition, the salinity increase with depth observed in Fig. 9a, also denote the influence of the river plume system at that location.

The evolution of the main plume forcing: river discharge and wind, were also analyzed since MA River plume seems to be the main feature promoting the different trends observed between coastal and oceanic locations. Tidal effects were discarded because tidal range is small in the shelf area ($\sim 20\text{--}40\text{ cm}$) (DiMarco and Reid, 1998; Salisbury et al., 2004) and because tidal components have shown a minimal contribution on MA plume development (Walker, 1996).

3.5. River discharge

River discharge of Mississippi and Atchafalaya Rivers presents an increase ($315\text{ m}^3\text{s}^{-1}\text{decade}^{-1}$), during the months under interest for the period 1994–2018, and a marked decrease ($-1571\text{ m}^3\text{s}^{-1}\text{decade}^{-1}$) during a more extended period (1982–2018) (see Table 3). On the other hand, the river discharge presented an increase in trend of $1432\text{ m}^3\text{s}^{-1}\text{decade}^{-1}$ and $717\text{ m}^3\text{s}^{-1}\text{decade}^{-1}$ during the preceding months (July–October) for the periods 1994–2018 and 1982–2018, respectively (Table 3). River discharge during this season (July–October) can significantly affect the river plume pattern

during November–February due to the freshwater retention time in this area (Zhang et al., 2012). The increase in the river discharge for the period 1994–2018 is coherent with the decrease observed in salinity and density patterns along the coastal shelf (Fig. 8b and c), as well as with the salinity decrease in upper layers observed in Fig. 9c at coastal locations. Furthermore, the correlation between the SST and the river discharge was calculated since river discharge is the main responsible for the plume formation. The highest correlation between the NDJF SST and the river discharge for the period 1982–2018 (Fig. 11a) was obtained when the river discharge was considered during the previous four months (JASO). It should be noted that it takes at least 3 months to accumulate the freshwater content from Atchafalaya and Mississippi Rivers on the shelf (Zhang et al., 2012). Negative correlations were observed in the area occupied by the river plume, with maximum values up to -0.4 along the Louisiana-Texas Shelf. Similar results were obtained for the period 1994–2018. Fig. 11b shows the JASO river discharge increase throughout the study period. The correlation is low due to several facts. On the one hand, other external factors like the mean temperature of the North Atlantic or the inter-annual variability of the weather conditions also affect the SST. On the other hand, the effect of river discharge on coastal temperature is lagged in several months, which also decreases the significance of the correlations.

3.6. Wind

A decrease in wind speed was detected in the area occupied by the river plume from November to February both for the period 1994–2018 (results not shown) and for the extended period (1982–2018) (Fig. 12). The easterly wind component, which prevails in the area for the season under study (Schiller et al., 2011), was analyzed along the cooling area over both periods (Table 4). An increase in the frequency of negative values in the zonal wind component was detected, especially for the period 1994–2018, which indicates that easterly winds became more frequent.

3.7. Circulation patterns

River discharge buoyancy forcing together with the predominant easterly winds promote a westward alongshore current, as was previously shown in Walker (1996) and Schiller et al. (2011). The evolution of this alongshore current has an important role on the coastal water dynamic of the area (Schiller et al., 2011; Androulidakis and Kourafalou, 2013; Androulidakis et al., 2019). Therefore, changes in the NDJF westward current along the Louisiana-Texas shelf for the period 2006–2018 respect to the period 1994–2006 were analyzed in Fig. 13a. In this way, positive (negative) values indicate the increase (decrease) of the westward current. Positive values were detected, which denote a reinforcement of the westward current probably induced by the increase of the rivers discharge together with the increase in the frequency of easterly winds. Moreover, changes in the associated freshwater transport were also analyzed to know more in depth the impact of the current variability on river plume movement (Fig. 13b). Positive values were detected, which denote an increment of the westward freshwater transport, according to the increase in river discharge and westward currents detected. The few negative values showed in Fig. 13a and b are probably related to the complex circulation pattern near the Mississippi bird-foot delta. In summary, the prevailing westward transport pattern along the Louisiana-Texas shelf from November to February has been strengthened.

3.8. Stratification

As was commented in the previous sections, both river discharge and wind forcing, and consequently freshwater transport, play a key role in the thermohaline variability along the coastal area. This is due to freshwater inputs and wind strongly influence the stability of the water

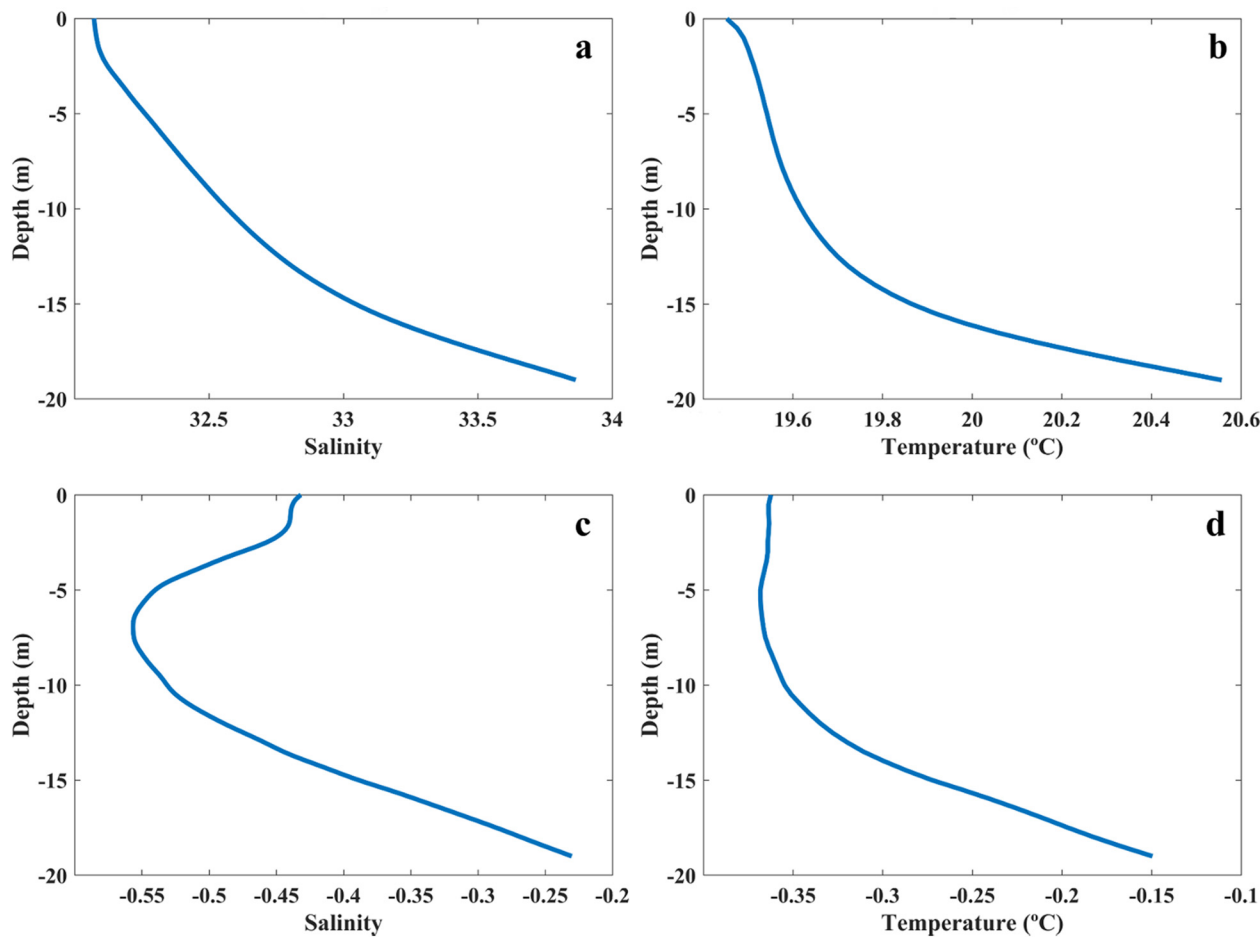


Fig. 9. Mean (a) NDJF salinity and (b) NDJF temperature vertical profiles in the coastal area over the period 1994–2018 using ROMS data. Changes in (c) NDJF salinity and (d) NDJF temperature calculated subtracting the mean values for the period 1994–2006 from the mean values for the period 2006–2018.

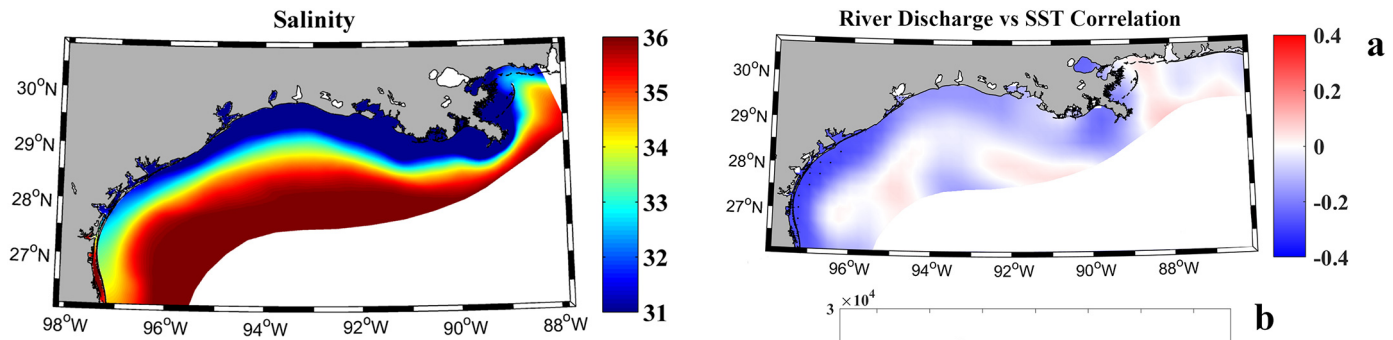


Fig. 10. Mean NDJF salinity calculated over the period 1994–2018 using ROMS data.

Table 3
Trends in river discharge. Statistical significance at 90% is marked with *.

		November–February	July–October
River runoff ($\text{m}^3\text{s}^{-1}\text{decade}^{-1}$)	1994–2018	315	1432*
	1982–2018	–1571*	717*

column (e.g. Kourafalou et al., 1996), conditioning the SST due to variations in stratification, and consequently, in the mixed layer depth and vertical mixing (Alexander et al., 2000). Changes in Brunt-Väisälä frequency (N^2) were calculated (Fig. 14) with the aim of detecting variations in the stratification patterns over the period 1994–2018.

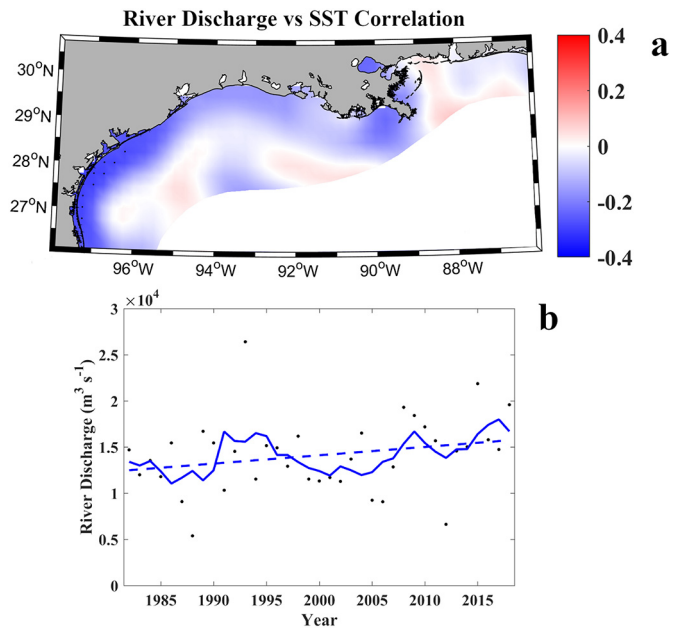


Fig. 11. (a) Spatial correlation between Mississippi-Atchafalaya River discharge (July–October) and SST from OISST (November–February) over the period 1982–2018. (b) Mississippi-Atchafalaya river discharge during the July–October season over the period 1982–2018. Points are raw data, solid line represents smoothed river discharge considering a range of ± 2 years, and the dashed line represents the linear trend.

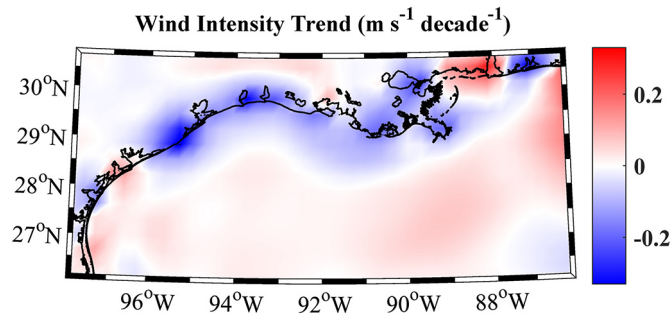


Fig. 12. Wind intensity trend ($\text{m s}^{-1} \text{ decade}^{-1}$) calculated using CFSR data from November to February over the period 1982–2018.

Table 4

Trends in the frequency of the zonal winds along the river plume area.

Parameter	Period	Trends (days decade ⁻¹)
East wind frequency trends	1994–2018	2.25
	1982–2018	0.91

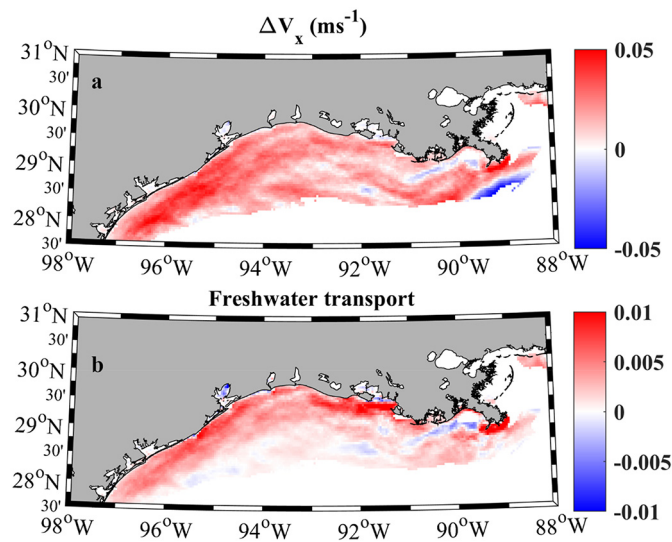


Fig. 13. Changes in the westward component of the (a) surface ocean current and (b) freshwater transport calculated using ROMS data. Mean values from November to February season from 1994 to 2006 were subtracted from mean values for the November to February season from 2006 to 2018. Positive values denote an increase in the westward movement.

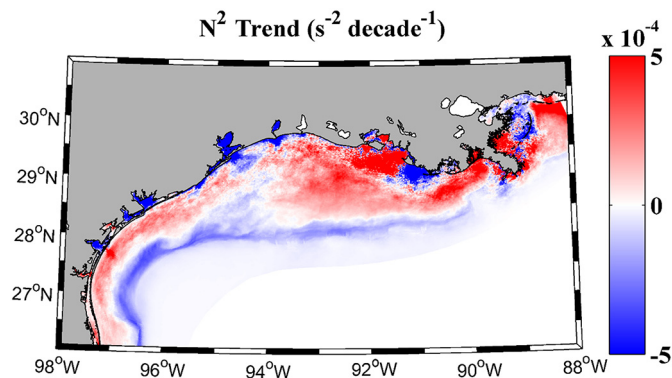


Fig. 14. Squared Brunt-Väisälä frequency (N^2) trend ($\text{s}^{-2} \text{ decade}^{-1}$) calculated using the ROMS data from November to February over the period 1994–2018.

Positive values, which denote an increase in the stratification, were only observed along the Louisiana-Texas shelf, in the area occupied by the river plume.

3.9. Empirical orthogonal function (EOF) analysis

The EOF analysis is often used in climate studies to analyze spatial patterns (modes) of variability and their temporal change. The EOF method decomposes the spatiotemporal variability into a set of orthogonal modes and the corresponding principal components (PCs) (Preisendorfer, 1988). These orthogonal modes can be identified, in our case, as oceanic modes, and therefore could help to explain the different SST patterns observed between coast and ocean. Each orthogonal mode is ranked by their temporal variance in such way that the sum of them reconstructs the initial signal. The EOFs analysis was applied to the NDJF SST anomalies over the period 1982–2018 (Fig. 15). Ocean and coastal points correspond to the locations marked with dots (magenta and blue respectively) in Fig. 4. The percentage of the NDJF SST variability explained by the two first EOFs modes (Fig. 15a and b) was 51.12% and 13.34% respectively, which constitute about 65% of the NDJF SST variability in the area. Each of the remaining EOFs modes explain < 10% of the NDJF SST variability and were discarded from the analysis. PCs of the leading EOFs modes were correlated with the NDJF SST at open ocean and near coast, the difference between them (NDJF ΔSST) defined as $\text{NDJF } \Delta\text{SST} = \text{NDJF } \text{SST}_{\text{ocean}} - \text{NDJF } \text{SST}_{\text{coast}}$, the AMO index and the river discharge. The first PC was mainly correlated with the NDJF SST at ocean locations (Fig. 15c) with $r = 0.96$ and a significance > 99% and with the AMO index at a less extent with $r = 0.36$ and a significance > 95%. The second PC was mainly correlated with the NDJF ΔSST (Fig. 15d) with $r = 0.83$ and a significance > 99% and also with the river discharge with $r = 0.30$ and a significance > 90%.

4. Discussion and conclusions

The Louisiana-Texas shelf shows a different SST pattern both in mean values (Fig. 5a) and trends (Figs. 2 and 3) from November to February over the period 1982–2018. This fact was corroborated by means of the EOF analysis, where the PCs of the leading modes showed to be correlated with coastal and oceanic SST anomalies (Fig. 15). First PC is mainly correlated with SST at ocean locations whereas the second PC is mainly correlated with ΔSST (SST difference between ocean and coast), which remarks that the temperature of both areas follows a different behavior. In addition, it was observed that the difference between coastal and oceanic waters has been reinforced during the last decade at a rate of $\sim 0.2^\circ\text{C year}^{-1}$, given rise to a difference on the order of 5°C at present (Fig. 5b). Previous studies (Wang et al., 1998; Walker et al., 2005) also reported a large cross-shelf SST gradient during winter months due to the fact that the Mississippi River is cooler than adjacent shelf waters and also due to the passage of cold fronts along the Northern Gulf of Mexico. To our knowledge, there are no previous studies that analyzed monthly SST trends considering the whole Louisiana-Texas shelf over periods longer than 30 years. Some studies (e.g. Li and Clarke, 2005; Muhling et al., 2012) analyzed the influence of SST variations on different fish populations along the area under scope. Glenn et al. (2015) studied SST trends in the Caribbean and surrounding region, including the whole Gulf of Mexico. Cooling trends in the Mississippi-Atchafalaya area were observed at their Fig. 3c, although they did not mention it explicitly. These authors found that the Atlantic Warm Pool, which flows from the Caribbean Sea into the Gulf of Mexico, has increased in size and intensity affecting the oceanic area considered in their study.

The mean area occupied by the plume (Fig. 10) coincides with the area where cooling was observed during winter months (Fig. 3). Therefore, the presence of the river plume is crucial to understand the different SST trends observed at coastal and oceanic locations. This is

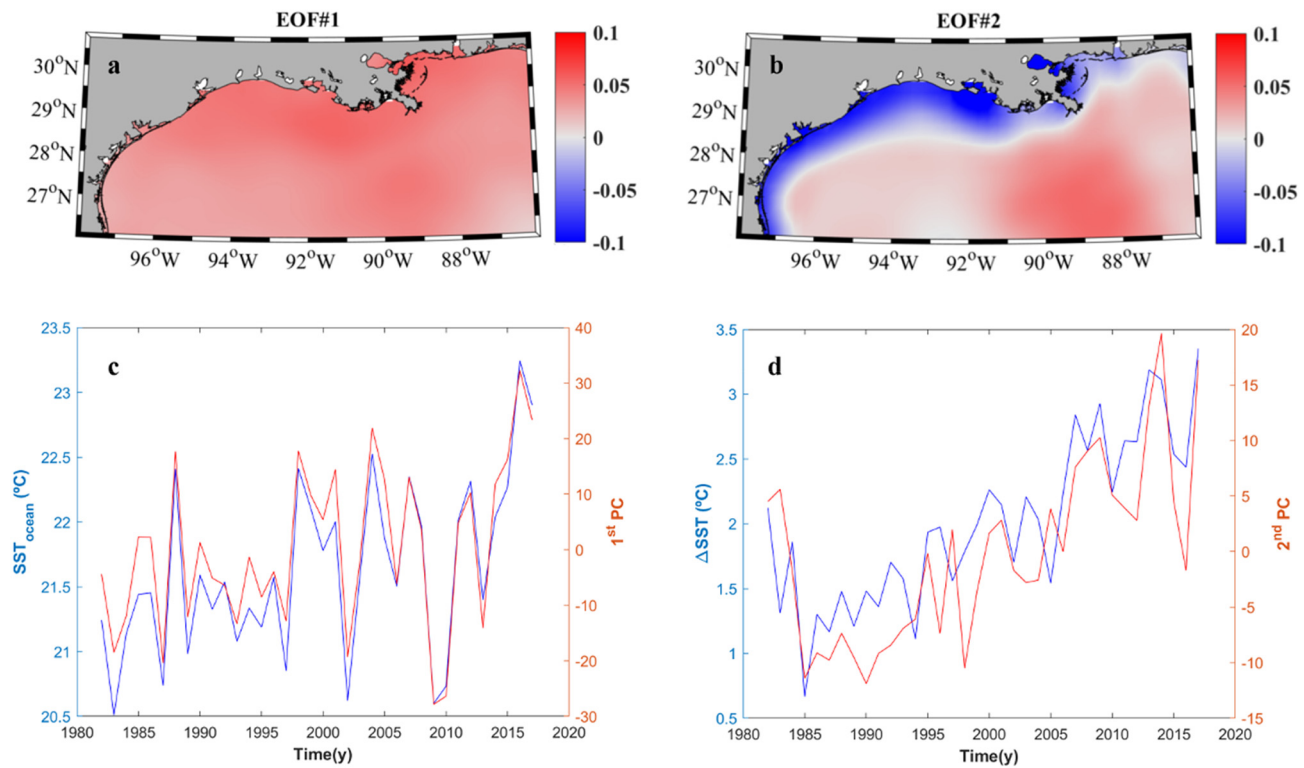


Fig. 15. EOF analysis carried out for NDJF SST anomalies over the period 1982–2018. (a) First and (b) second EOF modes. (c) NDJF SST_{ocean} vs PC#1 and (d) NDJF ΔSST vs PC#2. NDJF ΔSST = NDJF SST_{ocean} – NDJF SST_{coast}. (For interpretation of the references to color in this figure, the reader is referred to the web version of this article.)

also corroborated by analyzing the PCs of the EOFs. The AMO teleconnection index, which represents the SST anomaly of the North Atlantic, only shows significant correlation with the first PC (highly correlated with the oceanic NDJF SST). This fact reinforces the idea that whereas the oceanic part of the Gulf of Mexico follows a similar temperature behavior to that of the North Atlantic Ocean, the northwestern coastal area follows a different temperature pattern, being more dependent on coastal features. In fact, river discharge anomalies only show a significant correlation with the second PC (highly correlated with ΔSST), which shows that river flow is the main factor promoting the different behavior of the coastal area, since it is primarily responsible for river plume formation. Different drivers can be analyzed to better understand changes in the river plume and, hence, in coastal warming or cooling.

First of all, the trends in air-sea heat exchange (Fig. 6b) were spatially consistent with SST (Fig. 3), showing different patterns near shore and at the open ocean. Temperature decrease near coast is reflected in a declining rate of sea heat loss. On the other hand, temperature increase at ocean locations enhances ocean heat loss. In summary, heat loss evolution responds to the different warming trends detected, which denotes that changes in heat exchange are the consequence of SST changes not the cause. The air temperature showed a similar increase for the whole area including the continental locations (Fig. 7). This fact means that although this air warming can support the oceanic SST warming, it does not explain the coastal cooling.

Therefore, the cause behind the coastal cooling should be found in the ocean dynamics. With this purpose, changes in the stability of the water column were analyzed over the period 1994–2018. An increase in the buoyancy frequency was detected in the area occupied by the river plume (Fig. 14). This means that the higher stratification promoted by the haline forcing in the river plume area during winter months, is strengthening, which makes the vertical interchange of these waters even more difficult. Therefore, the cooling of surface water due to the entrainment of deeper waters is discarded, since, in addition deeper

water is warmer than surface water at this location (Fig. 9b).

The buoyancy gain can be mainly explained by the combination of two factors: an increase in Mississippi and Atchafalaya River discharges and a decrease in the wind intensity. Regarding river discharge, an increase about of $300 \text{ m}^3 \text{ s}^{-1} \text{ decade}^{-1}$ was detected considering both rivers from November–February months, and additionally, an increase about of $1400 \text{ m}^3 \text{ s}^{-1} \text{ decade}^{-1}$ was also observed in the preceding months (July–October). The fact of considering preceding months is important since it was estimated that at least 3 months are necessary to accumulate the freshwater content from Atchafalaya and Mississippi Rivers on the shelf (Zhang et al., 2012). The river discharge increase is corroborated by the respective salinity and density decrease at surface layers (Fig. 8b and c). In fact, a clear decrease in salinity is also observed in the first 10 m of the water column at river plume location (Fig. 9c). In addition, the wind intensity decrease in the river plume area, which also helps to reduce vertical mixing in seawater, reinforces the stratification. Thus, the increase of river discharge combined with the general decrease in wind intensity, support the idea of a buoyancy and a stratification increase along the Louisiana-Texas shelf from November to February over the period 1994–2018, which could favor the cooling of seawater by atmospheric interchange. This marked cooling is maintained in the first 10 m of the water column (Fig. 9d), which seems to be the depth highly influenced by the river plume since it is where the highest salinity decrease was detected (Fig. 9c). In addition, the prevailing westward alongshore current was reinforced (Fig. 13a), with an associated increase in the westward freshwater transport (Fig. 13b). This strengthening in the westward freshwater transport, which is mainly due to the increase both of the river discharge buoyancy and of the frequency of the easterly winds, limits the offshore spreading of plume waters reinforcing plume impact along the Louisiana-Texas shelf (Schiller et al., 2011; Androulidakis and Kourafalou, 2013; Androulidakis et al., 2019). In the case of the extended period, from 1982 to 2018, the increase in river discharge is only detected in the preceding months (July–October), which seems to play a key role in the

winter dynamics of the area, as was shown in previous studies (e.g. Zhang et al., 2012). In fact, this increase in river discharge significantly affects winter SST, as can be seen in the Fig. 11, where a negative correlation is detected between river discharge and SST. This means that the increase in river discharge promotes a SST decrease, probably due to the reinforcement of the stratification as was explained above. In addition, a marked decrease in wind intensity over the plume area is detected (Fig. 12), which also favors the stratification increase, as well as an increase in the easterly wind component. Although SST cooling in the plume area is detected regardless the time period considered, cooling is more evident since the 90s (Fig. 5), probably due to the large increase in river discharge, which seems to be the main responsible of the patterns obtained.

The influence of river plumes on SST has been previously analyzed in different regions worldwide (Howden and Murtugudde, 2001; Vizy and Cook, 2010; Park et al., 2011; Mignot et al., 2012; White and Toumi, 2014; Behara and Vinayachandran, 2016). These studies relate the presence of large river plumes with variations in the mixed layer depth and the stratification of the water column. The reinforcement of the river plume provokes a higher stratification, which in turn favors modifications in the ocean-atmosphere heat exchange. In this way, different studies analyzing large rivers during the warm season, such as: Congo River (Materia et al., 2012), Changjiang River (Park et al., 2011) or Amazonas/Orinoco Rivers (Vizy and Cook, 2010) found that the presence of freshwater inputs provokes a higher warming. This effect is mainly caused by the inhibition of both vertical mixing and entrainment of cold thermohaline water due to the higher stratification (Breugem et al., 2008). All these factors cause greater heat absorption in a shallow mixed layer. In addition, Behara and Vinayachandran (2016) propose the same mechanism to explain cooling in some areas of the Bay of Bengal during winter months. Thus, a stronger stratification provoked a more effective cooling over a thin mixed layer because ocean loses heat during winter months in this area. Costoya et al. (2016) also detected an increase in the rate of heat loss to explain cooling in the oceanic area affected by Loire and Gironde river plumes in the Bay of Biscay.

In the particular case of the Louisiana-Texas shelf, the impact of Mississippi-Atchafalaya River plume on SST trends is only remarkable during winter months, where cooling is observed in the coastal (plume) area whereas the adjacent ocean shows warming. During the rest of the year, there are not differences between plume and oceanic areas in terms of temperature evolution, following both a similar trend. Results obtained from the different analysis carried out in the present study support the idea that winter coastal cooling is based on the reinforcement of stratification along river plume waters. This is promoted by the buoyancy gain associated with the increase of river discharge and the wind evolution detected in the plume area with a decrease in its intensity. Trends of these forcing tend to reduce the vertical mixing and the entrainment of deeper water, which, in the plume area, is warmer than surface water due to the thermal inversion provoked by river discharge (Fig. 9b). This results in an enhancement of the atmospheric cooling in the river plume area. In addition, westward alongshore current and freshwater transport strengthen, insulating river plume from oceanic features and reinforcing alongshore plume impact (Schiller et al., 2011). Thus, the most probable cause of the different SST trend patterns along the Louisiana-Texas shelf during winter months, is the increase in the stratification of river plume waters, which leads to a more effective atmospheric cooling over there. This mechanism was previously proposed by Behara and Vinayachandran (2016) in the particular case of the Bay of Bengal. The cooling in the area affected by MA river plume is more intense during periods of plume strengthening (1994–2018), mainly driven by high river discharges, however, even when MA river plume does not experience great changes (1982–2018), its presence is able to modulate or even revert the adjacent ocean warming.

Future projections of water temperature indicate a water

temperature increase over the coming century (Bopp et al., 2013; Alexander et al., 2018; SROCC, 2019). These future projections remark the importance of the results obtained in this study. The knowledge and understanding of systems and mechanisms able to modulate at certain extent the future SST warming could be of a crucial importance, especially in key ecological and socio-economical areas such as the Louisiana-Texas shelf.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.jmarsys.2020.103374>.

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